

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

Duration : 1 Hour
Total Marks:50

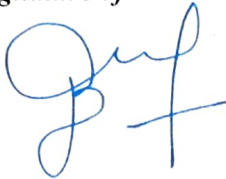
Code of Conduct:

(192511402)

I Bhuvaneshwaram (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

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1. A vacuum-cleaner robot operates in a two-room environment (Room A and Room B), each of which can be clean or dirty. Define the PEAS (Performance measure, Environment, Actuators, Sensors) description for this agent, and characterize its task environment along the dimensions of observability, determinism, episodicity, and number of agents.
2. A ride-hailing company operates a network of pickup points connected by roads with travel costs (time + fuel). Using the Uniform Cost Search (UCS) algorithm, determine the least-cost path from a starting point S to a destination G. Describe the frontier and explored set at each step for a graph of your choice with at least 6 nodes.
3. Explain Iterative Deepening Search (IDS) and trace its execution on a graph of your choice with at least 6 nodes, showing the search at depth limits 0, 1, and 2. Discuss why IDS combines the space-efficiency of DFS with the completeness of BFS.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem,	Understands problem with minor gaps	Partial understanding;	Misinterprets or unable to

		identifies all parameters and requirements accurately		misses key elements	understand problem
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① A vacuum-cleaner robot operates in a two room (A, B)
Define PRAS

* PRAS stands for

P- Performance, E- Environment, A- Actuators, S- Sensors

⇒ Performance Measures: clean both rooms, use minimum time and energy, avoid unnecessary movement

⇒ Environment: Two rooms (A and B) each can be clean or dirty

⇒ Actuators: Move left, Move right, Suck, Stop

⇒ Sensors: Detect current location, detect whether current room is clean or dirty.

* Task Environment characteristics:

* Observability:

Fully observable

The robot always knows

- Current room
- whether ~~room~~ is dirty or clean.

* Determinism:

A 'Suck' action always cleans the room

Same action → Same result.

* Episodic or Sequential:

cleaning Room A affects what happens next.
Current action influences future action.

* Number of Agents

only one robot is working

Property	Answer
Observability	Fully observable
Determinism	Deterministic
Episodic	Sequential
No. of agents	Single agent

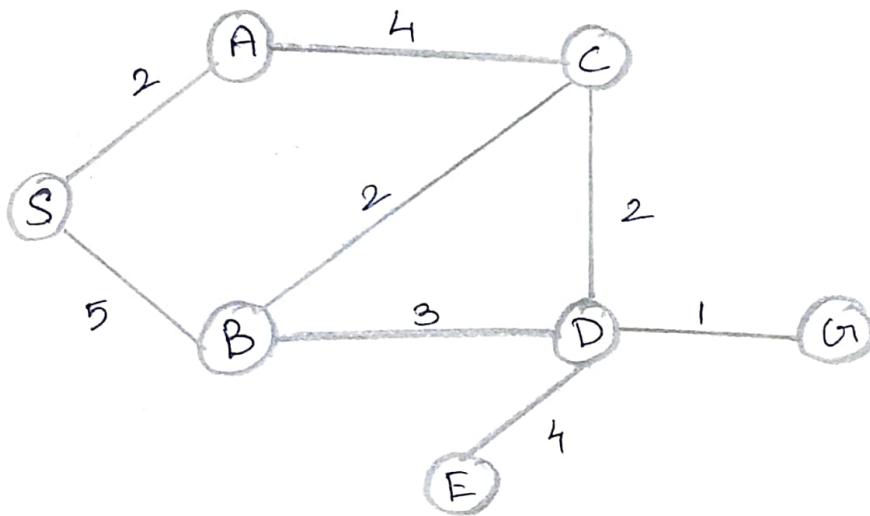
② A ride-hailing company operates a network of pickup pts connected by roads with travel costs. Using (VCS) algorithm

* Definition:

* Uniform cost search always expands the node with the lowest total path cost.

* It is similar to ~~Dijkstra's~~ Algorithm.
lowest cost first.

Example Graph



Goal:

$S \rightarrow G$

Step 3:

lowest cost

$B(5)$

explored
 S, A, B

New cost

$D=8$

$C(6)$

$D(8)$

Step 1:

$S(0)$

Expand S

$A(2)$

$B(5)$

Step 2

lowest cost

$A(2)$

~~$C(6)$~~

$B(5)$

$C(6)$

Step 4:

Expand

$C(6)$

Frontier

$D(8)$

Step 5:

Expand

$D(8)$

Add

$E(12)$

$G(9)$

Frontier

$G(9)$

$E(12)$

Step 6:

Expand

$G(9)$

Goal Reached

least cost path

$S \rightarrow B \rightarrow D \rightarrow G$

Total cost

$5 + 3 + 1 = 9$

Frontier	Explored
$S(0)$	S
$A(2), B(5)$	S
$B(5), C(6)$	S, A
$C(6), D(8)$	S, A, B
$D(8)$	S, A, B, C
$G(9), E(12)$	S, A, B, C, D

ucs Advantages:

Finds the least cost path

Complete and optimal

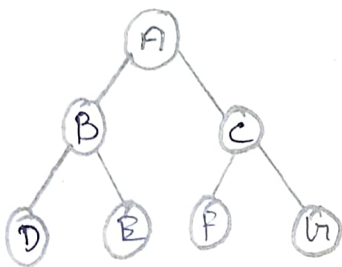
works with different path costs

Explain Iterative Deepening Search and trace its execution on a graph of your choice.

* Definition:

IDS repeatedly performs Depth Limited Search, increasing the depth limit each time until the goal is found.

Example Graph



Goal = F

Depth = 0

visit only root

A, not a goal

Depth = 1

visit A, B, C; not the goal

Depth = 2

visit A, B, D, E, C, F

Goal found

Search Table

Depth Limit	Nodes visited
0	A
1	ABC
2	ABDECF

* why DFS is space efficient?

DFS stores only one path

Memory:

$O(\text{depth})$

very small

* why DFS is Complete?

* It searches level by level like BFS

* Eventually reaches every depth

Therefore,

Goal is ~~always~~ found if it exists.